Physics 4B Spring 2026
Week 4 Discussion

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1. **Brainwarmer**[1]. A wall of an outdoor pool is $L = 10$ m long and $h = 2$ m deep. What force is needed to keep it in place?

   Solution: This question is about the water pressure on the wall. Recall that the excess pressure at a depth of y is $p(y) = \rho_w g y$. For a thin strip of the wall of thickness dy, the area is $dA = L\ dy$, so the force on the strip is

   $$F = \int p\ dA = \int_0^h L\ dy\ \rho_w g y = \frac{1}{2}\rho_w g L h^2 = 1.96 \times 10^5\ \text{N} \tag{1}$$

2. **Sound wave**. A loudspeaker produces a sinusoidal sound wave traveling through air. The wave has frequency $f = 800$ Hz, and the maximum excess pressure is $p_{\max} = 0.040$ Pa. Assume the density of air is $\rho_0 = 1.25$ kg/m^3 and the speed of sound is $v = 340$ m/s. Also assume the wave is a plane sinusoidal sound wave. Find:

   (a) The maximum displacement $s_{\max}$ of the air molecules.

   (b) The maximum change in density, $\Delta\rho_{\max}$.

   (c) The maximum speed of the air molecules.

   (d) At a certain location and instant, the excess pressure is $p = 0.020$ Pa and is increasing with time. At that instant, determine: the displacement s of the air at that location and the instantaneous change in density $\Delta\rho$.

   Solution:

   (a) The displacement as a function of x and t is $s(x,t) = s_{\max}\cos(kx - \omega t)$. From the derivation of the wave equation for sound waves in fluids, we know the relation[2]

   $$p = -B\frac{\partial s}{\partial x} \tag{2}$$

   so we get $p(x,t) = Bks_{\max}\sin(kx - \omega t)$. $p_{\max}$ is the amplitude of $p(x,t)$, so

   $$p_{\max} = Bks_{\max} = \rho_0\omega s_{\max} v \tag{3}$$

[1] Shankar 10.10

[2] If you need a reminder, this comes from the definition of B and expressing the change in volume in terms of the longitudinal displacement s.

where we used $B = \rho_0 v^2$. Therefore,

$$s_{\max} = \frac{p_{\max}}{\rho_0 \omega v} \approx 1.9 \times 10^{-8} \text{ m} \tag{4}$$

(b) We need to find a relation that relates the maximum change in density to what we already know. Recall that $B = \rho_0 v^2 = -\frac{p}{\Delta V/V}$. We can rewrite this into

$$p_{\max} = \frac{\rho_0 \Delta V_{\max}}{V} v^2 = \Delta\rho_{\max} v^2 \tag{5}$$

so we have

$$\Delta\rho_{\max} = \frac{p_{\max}}{v^2} \approx 3.5 \times 10^{-7} \text{ kg/m}^3 \tag{6}$$

Alternatively, we could have also used the relation $\Delta\rho = -\rho_0 \frac{\partial s}{\partial x}$, which is equivalent to what we did.

(c) To find the particle speed u, we take the time derivative of the displacement

$$u(x,t) = \omega s_{\max} \sin(kx - \omega t) \tag{7}$$

so the maximum particle speed is

$$u_{\max} = \omega s_{\max} \approx 9.4 \times 10^{-5} \text{ m/s} \tag{8}$$

(d) You can either use the new value for p and invert the expression $p(x,t)$ to find the value of $kx - \omega t$, and then plug that into $\Delta\rho(x,t)$ to find the new $\Delta\rho$, or you can do it more cleverly by using scaling arguments. From previous parts, we know that $p = \rho_0 \omega v s \propto s$ and $p = v^2 \Delta\rho \propto \Delta\rho$. From these relationships, we have

$$\frac{\Delta\rho}{\Delta\rho_{\max}} = \frac{p}{p_{\max}} = \frac{1}{2} \tag{9}$$

so $\Delta\rho \approx 1.7 \times 10^{-7}$ kg/m^3.

We can do this for $\Delta\rho$ because it is in phase with p. However, we cannot do the same for s because there is a phase shift of $\pi/2$ between $s(x,t)$ and $p(x,t)$, so we cannot do the simple scaling argument and have to use the "dumb" method.

Now, let $\phi = kx - \omega t$. Using $p/p_{\max} = 1/2$, we have $\sin\phi = 1/2$. Our goal is to figure out $\cos\phi$. There are two choices for ϕ, $\pi/6$ and $5\pi/6$. The problem says $p \propto \sin\phi$ is increasing **over time**, so ϕ has to be $5\pi/6$, and therefore $s = s_{\max}\cos\phi = -\frac{\sqrt{3}}{2} s_{\max} \approx -1.6 \times 10^8$ m.

3. **More sound wave**. A plane sinusoidal sound wave travels through air in a hollow tube of cross-sectional area $A = 12$ cm^2. The air density is 1.2 kg/m^3 and the sound speed is $v = 343$ m/s. The wave has frequency $f = 600$ Hz and maximum excess pressure $p_{\max} = 0.050$ Pa. Consider a short segment of air inside the tube of length $\Delta x = 0.50$ cm. Assume the segment is short enough that the pressure gradient across it can be treated as approximately constant.

   (a) At an instant when the pressure inside this segment is maximum, what is the net force acting on the air in the segment?

   (b) At an instant when the excess pressure in the segment is zero, what are the possible values of the net force on the segment?

   (c) For part (b), what are the possible values of the acceleration of the air in that segment at that instant?

   (d) Why must the segment length Δx be "small"? Small compared to what? Explain what goes wrong if the segment is not small.

   Solution:

   (a) The previous problem is related to the derivation of the $\partial s/\partial x$ side of the wave equation. This problem is related to the $\partial s/\partial t$ side.

   The net force on the fluid segment is

$$F_{\text{net}} = A(p(x) - p(x + \Delta x)) = -A\frac{\partial p}{\partial x}\Delta x \tag{10}$$

   From the previous problem we know that the excess pressure as a function of x and t is $p(x,t) = Bks_{\max}\sin(kx - \omega t)$, so

$$|F_{\text{net}}|_{\max} = A\Delta x Bk^2 s_{\max} = A\Delta x(\rho v^2)\frac{\omega^2}{v^2}\frac{p_{\max}}{\rho\omega v} = A\Delta x p_{\max}\frac{\omega}{v} \approx 3.3\times 10^{-6}\text{ N} \tag{11}$$

   However, this is the maximum possible force, not the net force when the pressure is at maximum, because there is a phase shift between the two! When the pressure is at maximum, its derivative is 0, so the net force is actually 0!

   (b) When $p = 0$, the derivative of $p(x,t)$ is at maximum or minimum, so the answer here is $F_{\text{net}} = \pm 3.3\times 10^{-6}$ N.

   (c) The acceleration of air particles would be[3] $\ddot{s}(x,t) = -\omega^2 s_0\cos(kx - \omega t)$. At the instant in part (b), the force is at maximum/minimum, so the magnitude of the acceleration has to be at maximum as well. Therefore, $a = \pm\omega^2 s_{\max} = \pm\omega\frac{p_{\max}}{\rho v} = \pm 0.46$ m/s^2.

[3] Alternatively, we can use $F_{\text{net}} = m\ddot{s} = \rho A\Delta x\ddot{s}$.

(d) The length of the segment must be small so that our approximation

$$p(x+\Delta) - p(x) \approx \frac{\partial p}{\partial x}\Delta x \tag{12}$$

is good. This ensures that the pressure doesn't change too much from one end of the segment to the other end, so that $\partial p/\partial x$ is roughly a constant over the segment.

The length Δx must be small compared to the wavelength, which characterizes the separation between segments of noticeably different pressures. We want to Δx to describe a chunk of fluid with roughly uniform pressure, if $\Delta x \not\ll \lambda$, the approximation we used would fail.